PEARSON EDEXCEL INTERNATIONAL A LEVEL · CHEMISTRY · WCH11

03

Bonding and Structure

Ionic, covalent and metallic — the forces that decide how everything behaves.

Ionic, covalent and metallic bonding; dot-and-cross diagrams; shapes of molecules by VSEPR; polarity; and how bonding explains physical properties.

UNIT	SECTIONS	STUDY TIME	FLASHCARDS
1 · WCH11	23	~6.5 h	27

COVERS **23 SPECIFICATION POINTS** — STATEMENTS QUOTED VERBATIM FROM THE SPECIFICATION

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HOW TO USE THIS DOCUMENT

Every section opens with the specification statement, worded exactly as Edexcel words it — that is what the examiner marks against. Definitions, equations and exam alerts are boxed; worked examples walk the full method step by step. Read a section, shut the page and recall its bullets from memory, then grind the topic's flashcard deck on the site.

SPEC 3.1 - 3.22 ≈ 10 MIN

Specification Map

THE SPECIFICATION SAYS

Students will be assessed on their ability to:

Every statement the specification assesses in this topic — use it as your revision checklist.

SPEC	ASSESSMENT STATEMENT
3.1	know and be able to interpret evidence for the existence of ions, limited to physical properties of ionic compounds, electron density maps and the migration of ions
3.2	be able to describe the formation of ions in terms of loss or gain of electrons
3.3	be able to draw dot-and-cross diagrams to show electrons in cations and anions
3.4	be able to describe ionic crystals as giant lattices of ions
3.5	know that ionic bonding is the result of strong net electrostatic attraction between ions
3.6	understand the effects of ionic radius and ionic charge on the strength of ionic bonding
3.7	understand reasons for the trends in ionic radii down a group in the Periodic Table, and for a set of isoelectronic ions, including N^{3-} to Al^{3+}
3.8	understand the meaning of the term ‘polarisation’ as applied to ions
3.9	understand that the polarising power of a cation depends on its radius and charge, and the polarisability of an anion also depends on its radius and charge
3.10	understand that covalent bonding is the strong electrostatic attraction between two nuclei and the shared pair of electrons between them, based on the evidence: i the physical properties of giant atomic structures ii electron density maps for simple molecules
3.11	be able to draw dot-and-cross diagrams to show electrons in covalent substances, including: i molecules with single, double and triple bonds ii species with dative covalent (coordinate) bonds, including Al_2Cl_6 and the ammonium ion
3.12	be able to describe the different structures formed by giant lattices of carbon atoms, including graphite, diamond and graphene, and discuss the applications of each
3.13	understand the meaning of the term ‘electronegativity’ as applied to atoms in a covalent bond
3.14	know that ionic and covalent bonding are the extremes of a continuum of bonding type and be able to explain this in terms of electronegativity differences, leading to bond polarity in bonds and molecules, and to ionic bonding if the electronegativity is large enough
3.15	be able to distinguish between polar bonds and polar molecules and predict whether or not a given molecule is likely to be polar
3.16	understand the principles of the electron-pair repulsion theory, used to interpret and predict the shapes of simple molecules and ions

SPEC	ASSESSMENT STATEMENT
3.17	understand the terms ‘bond length’ and ‘bond angle’
3.18	know and be able to explain the shapes of, and bond angles in, $BeCl_2$, BCl_3, CH_4, NH_3, NH_4^+, H_2O, CO_2, gaseous PCl_5, SF_6 and C_2H_4
3.19	be able to apply the electron-pair repulsion theory to predict the shapes of, and bond angles in, molecules and ions analogous to those in 3.18
3.20	understand that metals consist of giant lattices of metal ions in a sea of delocalised electrons
3.21	know that metallic bonding is the strong electrostatic attraction between metal ions and the delocalised electrons
3.22	be able to use the models in 3.20 and 3.21 to interpret simple properties of metals, including electrical conductivity and high melting temperature

SPEC 3.1 ≈ 20 MIN

Evidence for Ions

> THE SPECIFICATION SAYS
>
> **know and be able to interpret evidence for the existence of ions, limited to physical properties of ionic compounds, electron density maps and the migration of ions**

Physical properties, electron density maps and migrating ions — three independent proofs.

Ions cannot be seen directly, so the evidence for their existence is indirect — and the specification limits it to three lines of attack: the physical properties of ionic compounds, electron density maps and the migration of ions. Each points to the same conclusion: charged particles are locked in a lattice when solid, and free to move when molten or dissolved.

Evidence 1 — the physical properties of ionic compounds

- **High melting and boiling temperatures** indicate strong electrostatic forces acting throughout a three-dimensional lattice — a great deal of energy is needed to pull the ions apart.
- Solid ionic compounds **do not conduct electricity**: the ions are held in fixed positions in the lattice, so there are no mobile charged particles to carry the current.
- The same compound **conducts well when molten or in aqueous solution** — melting or dissolving frees the ions to move.
- Taken together, these properties prove the existence of **charged particles (ions)** that are fixed in a lattice when solid but mobile when molten or dissolved.

Evidence 2 — electron density maps

- Electron density maps (obtained from **X-ray diffraction** experiments) show **discrete, spherical regions of high electron density** — the ions themselves.

- Between a cation and an adjacent anion there is a **region of zero electron density** — the density does not bridge the gap between the two nuclei.
- This is the signature of **complete electron transfer**: the electrons belong to one ion or the other, rather than being shared between the nuclei as they are in a covalent bond.

TIP

Compare with the covalent maps in 3.10

In an ionic electron density map the density **falls to zero between** the cation and the anion. In a covalent map (Cl_2, HCl) the density is **highest between** the two nuclei. Same technique, opposite conclusion.

Evidence 3 — the migration of ions

- Place a green solution of **copper(II) chromate(VI), $CuCrO_4$**, on moist filter paper (or in a U-tube) and apply a potential difference between two electrodes.
- A **blue colour moves towards the cathode** (the negative electrode) — carried by the blue **Cu^{2+} cations**, which are attracted to the negative electrode.
- A **yellow colour moves towards the anode** (the positive electrode) — carried by the yellow **CrO_4^{2-} anions**, which are attracted to the positive electrode.
- The direction each coloured ion travels reveals its **charge** — and the fact that the colours move at all proves that **ions exist and are mobile** in solution.

$$CuCrO_4(aq) \rightarrow Cu^{2+}(aq) + CrO_4^{2-}(aq)$$

Dissociation of copper(II) chromate(VI) — blue cation, yellow anion

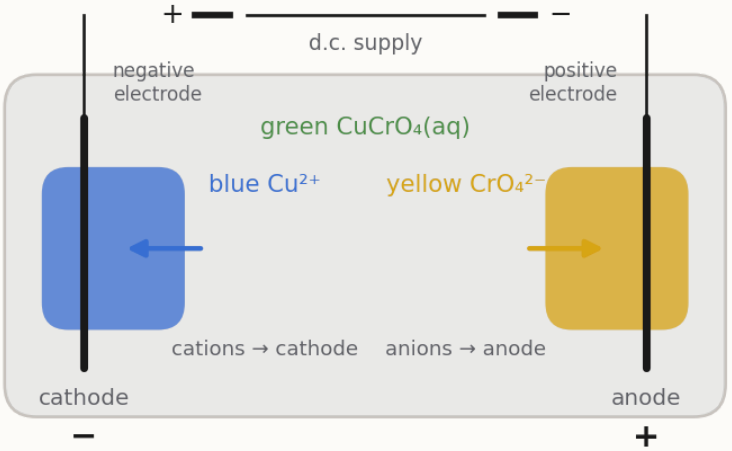


FIG — *The migration of ions — blue Cu^{2+} moves towards the negative electrode while yellow CrO_4^{2-} moves towards the positive electrode.*

- The same evidence can be obtained with **potassium manganate(VII)** crystals on a moist filter paper or microscope slide: the purple **MnO_4^- anions** migrate towards the **anode**.

EXAM ALERT

Interpret, don't just recall

The spec says **interpret** the evidence — so expect an observation and a conclusion to draw. Name the moving species, state its charge from the electrode it heads for, and finish with the conclusion: the particles are **ions**.

SPEC 3.2 ≈ 15 MIN

Formation of Ions

THE SPECIFICATION SAYS

be able to describe the formation of ions in terms of loss or gain of electrons

Metals lose, non-metals gain — the octet rule at work.

DEFINITION

Cation

A positively charged ion, formed when an atom **loses** one or more of its outer-shell (valence) electrons.

DEFINITION

Anion

A negatively charged ion, formed when an atom **gains** one or more electrons into its outer shell.

- **Metal atoms lose** their outer-shell electrons to achieve a stable **noble gas electronic configuration** — with more protons than electrons, the particle left behind is positive.
- **Non-metal atoms gain** electrons into their outer shell to complete a full **octet** (eight outer electrons) — with more electrons than protons, the particle formed is negative.
- The **nucleus never changes** during ion formation — an ion still has the same number of protons as the atom, so it is still the same element.
- The **charge on the ion equals the number of electrons lost or gained**.

Writing the equations

$$Na \rightarrow Na^+ + e^-$$

Sodium (2,8,1) loses its single outer electron → 2,8

$$Mg \rightarrow Mg^{2+} + 2e^-$$

Magnesium (2,8,2) loses two outer electrons → 2,8

$$O + 2e^- \rightarrow O^{2-}$$

Oxygen (2,6) gains two electrons → 2,8

$$Cl + e^- \rightarrow Cl^-$$

Chlorine (2,8,7) gains one electron → 2,8,8

- The charge follows the **Group number**: Group 1 → +1, Group 2 → +2, Group 13 → +3; Group 15 → 3−, Group 16 → 2−, Group 17 → 1−.
- Transition metals form **more than one ion** (e.g. Fe^{2+}/Fe^{3+}, Cu^{+}/Cu^{2+}) — the charge is then shown in the name, as in iron(III) chloride.

KEY POINT

Ions have noble gas configurations

Na^+ has the electron arrangement **2,8** (like neon); Cl^- has **2,8,8** (like argon). The drive towards a noble gas configuration is what makes the ion stable.

SPEC 3.3 ≈ 20 MIN

Dot-and-Cross Diagrams for Ions

THE SPECIFICATION SAYS

be able to draw dot-and-cross diagrams to show electrons in cations and anions

Square brackets, charges and transferred electrons — drawn the mark-scheme way.

Rules for ionic dot-and-cross diagrams

1. Draw **outer-shell electrons only**, unless the question asks for all shells.
2. Put **square brackets** around each ion with the **overall charge at the top right** (e.g. $[Na]^+$, $[Cl]^-$).
3. Use **different symbols for different atoms** — dots for one, crosses for the other — so the **transferred electrons** can be traced back to the metal atom that donated them.
4. Make sure the **electron count in each ion is correct** — the anion's outer shell is always full (8 electrons).

WORKED EXAMPLE **Sodium chloride, NaCl**

PROBLEM

Draw the dot-and-cross diagram for sodium chloride.

1. Sodium (2,8,1) has **one outer electron**; chlorine (2,8,7) has **seven**.
2. Transfer **one electron** from sodium to chlorine: draw the transferred electron as a **cross** inside chlorine's outer shell.
3. Enclose each ion in **square brackets** with its charge: $[Na]^+$ and $[Cl]^-$.

ANSWER **Na^+ with a full outer shell (2,8) and Cl^- with a full outer shell (2,8,8 — seven dots and one cross), each in square brackets with the charge shown.**

WORKED EXAMPLE **Magnesium oxide, MgO**

PROBLEM

Draw the dot-and-cross diagram for magnesium oxide.

1. Magnesium (2,8,2) must **lose two electrons**; oxygen (2,8,6) must **gain two**.
2. Transfer **two electrons** from magnesium to oxygen — both are drawn as crosses in oxygen's outer shell.
3. The charges are **2+** and **2−** — one magnesium atom supplies one oxygen atom.

ANSWER **$[Mg]^{2+}$ and $[O]^{2-}$, with the two transferred electrons shown as crosses inside oxygen's shell.**

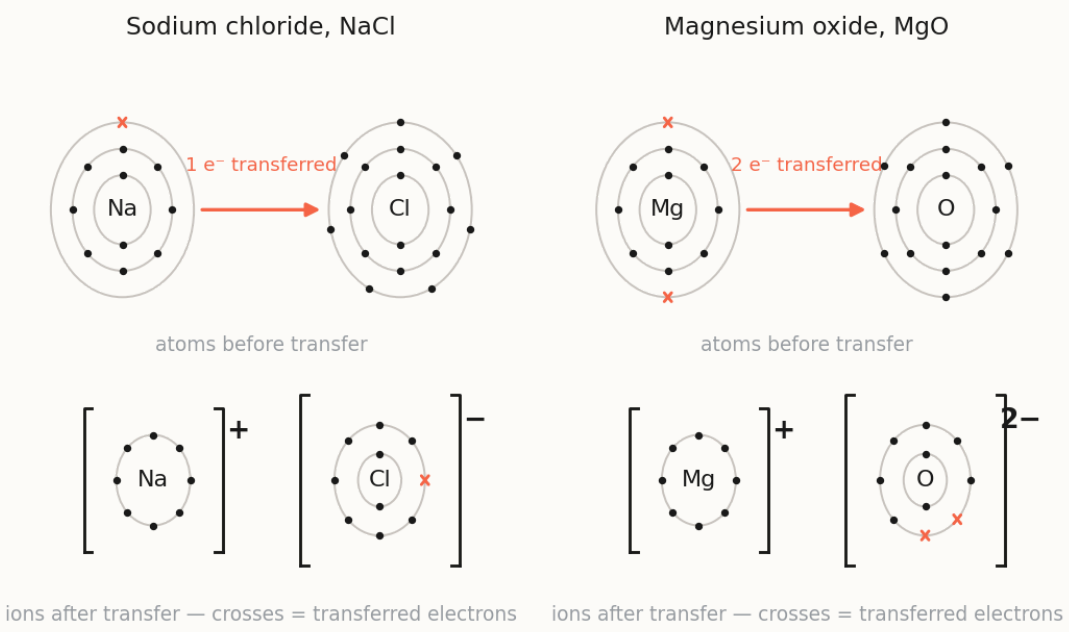


FIG — *Electron transfer drawn the mark-scheme way — outer shells only, dots and crosses, square brackets with charges, and the transferred electrons visible inside the anion.*

EXAM ALERT

The formula follows from the charges

In sodium oxide, **two** sodium atoms each donate one electron to the **same** oxygen atom — so the formula is Na_2O. Total electrons lost must equal total electrons gained: the compound is electrically neutral.

SPEC 3.4 ≈ 20 MIN

Giant Ionic Lattices

THE SPECIFICATION SAYS

be able to describe ionic crystals as giant lattices of ions

Alternating ions in a regular 3D array — 6:6 coordination in sodium chloride.

DEFINITION

Giant ionic lattice

A regular three-dimensional repeating array of alternating positive cations and negative anions, held together by strong electrostatic attraction extending in all directions.

- The ions sit in a **regular, repeating 3D pattern** — the arrangement repeats through millions of ions in every direction, so ionic crystals are **giant** structures.

- The electrostatic attraction is **non-directional**: each ion attracts oppositely charged ions **all around it**, which is what builds the lattice.
- There are **no discrete molecules** in an ionic crystal — a formula such as NaCl is the **simplest ratio** of ions (an empirical formula), not a molecule.
- The ratio keeps the crystal **electrically neutral** overall — equal amounts of positive and negative charge.

Coordination number in sodium chloride

DEFINITION

Coordination number

The number of nearest-neighbour ions of **opposite charge** surrounding a given ion in a lattice.

- In NaCl, each **Na^+ is surrounded octahedrally by 6 Cl^-**, and each **Cl^- is surrounded by 6 Na^+** — described as **6:6 coordination**.
- This arrangement lets every cation sit close to several anions, so the attractions on every ion outweigh its repulsions.

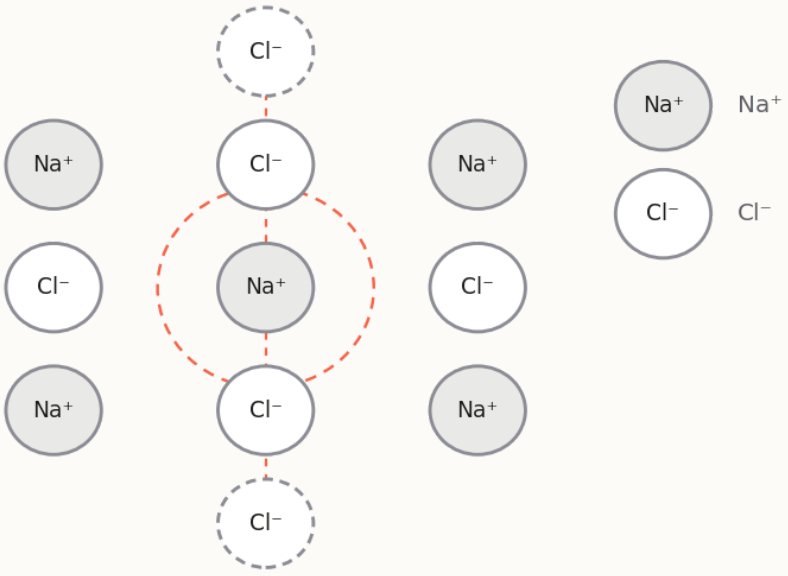


FIG — *The sodium chloride lattice — alternating Na^+ and Cl^- in all three dimensions. The highlighted Na^+ touches six Cl^- ions: 6:6 coordination.*

SPEC 3.5 ≈ 10 MIN

Ionic Bonding

THE SPECIFICATION SAYS

know that ionic bonding is the result of strong net electrostatic attraction between ions

Strong net electrostatic attraction — the attractions outweigh the repulsions.

DEFINITION

Ionic bonding

The strong **net** electrostatic attraction between oppositely charged ions in a giant lattice.

- In the lattice, **attractions between oppositely charged ions** act alongside **repulsions between ions of like charge**.
- The attractions **outweigh** the repulsions — the balance is a **net attraction**, and it is this net attraction that holds the lattice together.
- Because the attraction is **non-directional**, the bonding acts **throughout the whole lattice**, in every direction — this is what makes the compound 'giant'.
- This one force explains the properties met in 3.1: high melting temperatures (much energy needed to overcome the attraction) and conduction only when the ions are free to move.

FORCES INSIDE THE LATTICE	BETWEEN	RELATIVE EFFECT
Attractive forces	Each cation and the surrounding anions	Dominate — hold the lattice together
Repulsive forces	Cation–cation and anion–anion neighbours	Present, but outweighed by the attractions

KEY POINT

The word 'net' earns the mark

Definition questions want 'strong **net** electrostatic attraction between oppositely charged ions' — the 'net' acknowledges that repulsions exist but are outweighed.

SPEC 3.6 ≈ 20 MIN

Ionic Radius, Charge and Bond Strength

THE SPECIFICATION SAYS

understand the effects of ionic radius and ionic charge on the strength of ionic bonding

Why MgO out-melts NaCl — charge up, radius down.

Two factors decide how strongly the ions in a lattice attract each other — and therefore how much energy it takes to melt the crystal.

- **Higher ionic charge** — ions with larger charges exert a significantly stronger electrostatic attraction on each other (Mg^{2+} attracts more strongly than Na^{+}; O^{2-} more than F^{-}).
- **Smaller ionic radius** — smaller ions allow the **centres of charge to sit closer together**, and electrostatic attraction grows rapidly as the separation falls.
- The two factors combine into one idea: **charge density** — charge per unit volume. A small, highly charged ion packs a lot of charge into a small space.

WORKED EXAMPLE **Why does MgO melt over 2000 °C higher than NaCl?**

PROBLEM

Magnesium oxide melts at 2852 °C; sodium chloride melts at 801 °C. Explain the difference.

1. Mg^{2+} and O^{2-} are **doubly charged**; Na^{+} and Cl^{-} are singly charged — so the attractions in MgO are much stronger.
2. Mg^{2+} (72 pm) is **smaller** than Na^{+} (102 pm), and O^{2-} (140 pm) is **smaller** than Cl^{-} (181 pm) — the centres of charge in MgO sit closer together.
3. Both effects point the same way: much stronger ionic bonding in MgO.

ANSWER **MgO's ions are more highly charged and smaller than NaCl's, so much more energy is needed to melt it (2852 °C vs 801 °C).**

COMPOUND	IONS	MELTING POINT	WHY
NaCl	Na^{+} and Cl^{-} — both singly charged	801 °C	Baseline ionic lattice
NaF	Na^{+} and F^{-} — same charges as NaCl	993 °C	F^{-} is smaller than Cl^{-}, so the attraction is stronger
MgO	Mg^{2+} and O^{2-} — both doubly charged	2852 °C	Higher charges and smaller ions — much stronger bonding

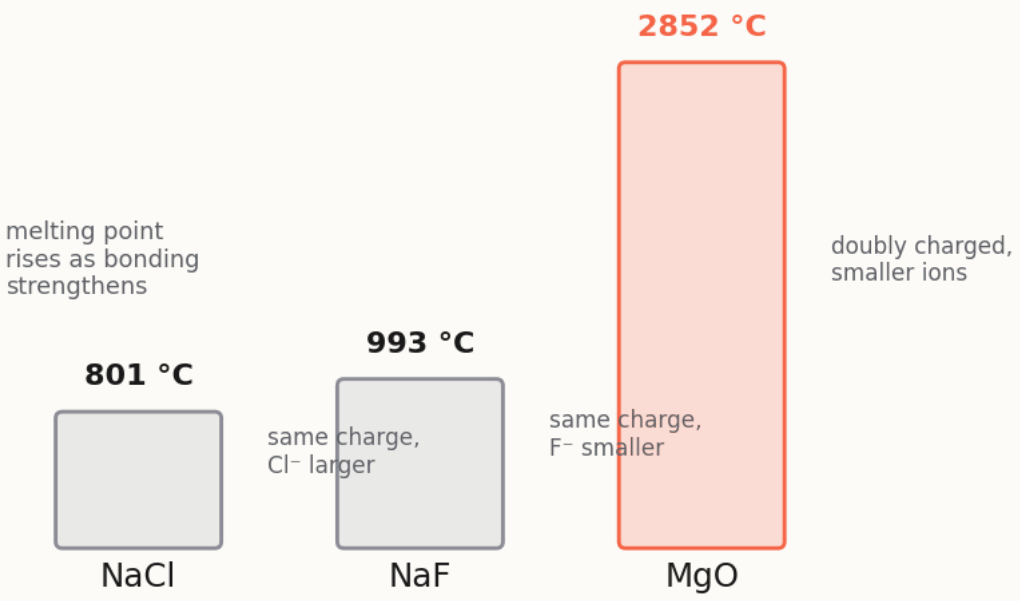


FIG – *Melting points climb as the charges rise and the radii fall — NaCl 801 °C, NaF 993 °C, MgO 2852 °C.*

TIP

Read melting points as bond strength

For ionic compounds the melting point is a measure of the **strength of the ionic bonding** — the higher the melting point, the stronger the net electrostatic attraction in the lattice.

SPEC 3.7 ≈ 20 MIN

Trends in Ionic Radii

THE SPECIFICATION SAYS

understand reasons for the trends in ionic radii down a group in the Periodic Table, and for a set of isoelectronic ions, including N^{3-} to Al^{3+}

Down a group, and across an isoelectronic series from N^{3-} to Al^{3+}.

DEFINITION

Isoelectronic

Describes species that share the **same number of electrons** — and therefore the same electronic configuration.

Down a group

- Ionic radius **increases** down any group, because **extra quantum shells** of electrons are added with each new period.

- The outer electrons sit in shells further and further from the nucleus — and the inner shells **shield** the outer electrons from the nuclear charge.

Across an isoelectronic series: N^{3-} to Al^{3+}

- The ions **N^{3-}, O^{2-}, F^-, Na^+, Mg^{2+}, Al^{3+}** all contain **10 electrons** (configuration $1s^2\ 2s^2\ 2p^6$) — they are isoelectronic.
- The **proton number rises** from N (7 protons) to Al (13 protons) across the series.
- The **same number of electrons** is pulled by an **increasing nuclear charge**, so the electron cloud is drawn in more and more tightly.
- Therefore the ionic radius **decreases** from N^{3-} (the largest) to Al^{3+} (the smallest).

ION	PROTONS	ELECTRONS	RADIUS (PM)
N^{3-}	7	10	171
O^{2-}	8	10	140
F^-	9	10	133
Na^+	11	10	102
Mg^{2+}	12	10	72
Al^{3+}	13	10	54

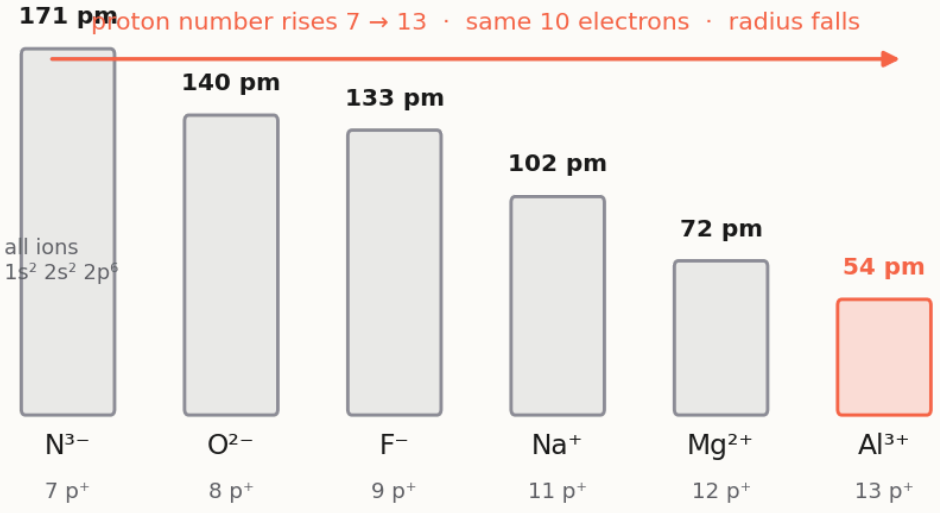


FIG — *One electron cloud (10 electrons), six different nuclei — as the proton count climbs from 7 to 13, the radius falls from 171 pm to 54 pm.*

- Compare ions with their parent atoms: a **cation is smaller** than its atom (its outer shell is lost), while an **anion is larger** (the extra electron increases repulsion within the shell).

EXAM ALERT

Same electrons? Count the protons

Isoelectronic questions are solved with one line: **same number of electrons, more protons → smaller ion**. Order the species by proton number and you have the size order immediately.

SPEC 3.8 ≈ 15 MIN

Polarisation of Ions

THE SPECIFICATION SAYS

understand the meaning of the term 'polarisation' as applied to ions

When a cation distorts an anion — and the bond gains covalent character.

DEFINITION

Polarisation (of an anion)

The distortion of the electron density cloud of an anion caused by the electrostatic attraction of a neighbouring cation.

- In a perfectly ionic compound, each ion is an **undistorted sphere** of electron density — complete electron transfer, no sharing.
- In reality, a neighbouring **cation pulls on the anion's outer electrons**, dragging the anion's cloud towards itself.
- When the distortion is significant, electron density is pulled **into the region between the two nuclei** — precisely where the shared density of a covalent bond sits.
- The bond therefore gains **covalent character**: pure ionic and pure covalent bonding are the **extremes of a continuum** (see 3.14), and polarisation is what shifts a bond along it.

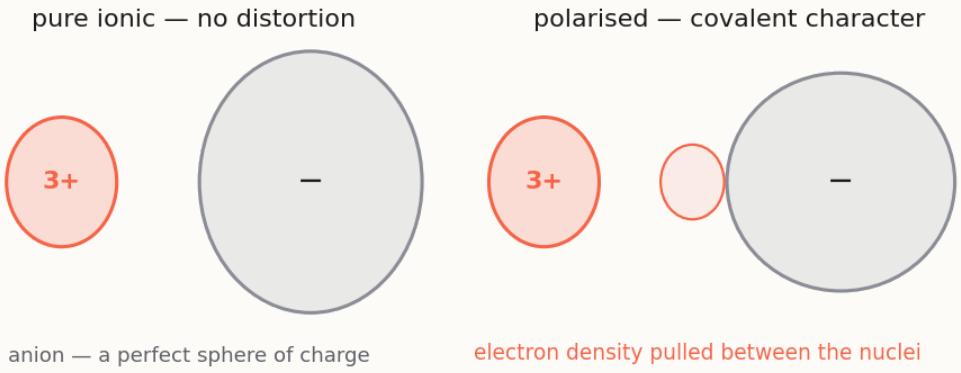


FIG – *Polarisation — the cation's attraction drags the anion's electron cloud towards the region between the nuclei, introducing covalent character.*

TIP

Which compounds show it most?

Look for a **small, highly charged cation** paired with a **large, highly charged anion** — e.g. $AlCl_3$ and Al_2O_3 show far more covalent character than NaCl. Section 3.9 turns this observation into a rule.

SPEC 3.9 ≈ 15 MIN

Polarising Power and Polarisability

THE SPECIFICATION SAYS

understand that the polarising power of a cation depends on its radius and charge, and the polarisability of an anion also depends on its radius and charge

Small, highly charged cations distort large, highly charged anions.

DEFINITION

Polarising power (of a cation)

The ability of a cation to distort the electron cloud of a neighbouring anion — highest when the cation has a **high charge and a small ionic radius** (high charge density).

DEFINITION

Polarisability (of an anion)

How easily an anion's electron cloud is distorted — highest when the anion has a **large radius and a high negative charge**.

- **Cations:** polarising power rises with **charge** ($Al^{3+} > Mg^{2+} > Na^+$) and with **smaller radius** — Li^+ is unusually polarising for a 1+ ion.
- **Li^+ and Al^{3+}** are the classic high polarising power cations — small, with their charge concentrated in a small volume.
- **Anions:** polarisability rises with **radius** ($I^- > Br^- > Cl^- > F^-$) and with **more negative charge**.
- **I^- is much more polarisable than F^-**: its outer electrons are further from the nucleus and heavily shielded, so its electron cloud distorts easily.

WORKED EXAMPLE **Why does aluminium chloride show covalent character?**

PROBLEM

Aluminium is a metal and chlorine a non-metal, yet $AlCl_3$ behaves largely as a covalent, molecular substance. Explain why.

1. The **Al^{3+} cation is very small and carries a 3+ charge** — extremely high polarising power.
2. The **Cl^- anion is large** and therefore easily polarisable.
3. The chloride's electron cloud is pulled strongly into the region between the two nuclei.

ANSWER **Significant polarisation introduces substantial covalent character — which is why $AlCl_3$ is molecular (and dimerises to Al_2Cl_6, see 3.11) rather than truly ionic.**

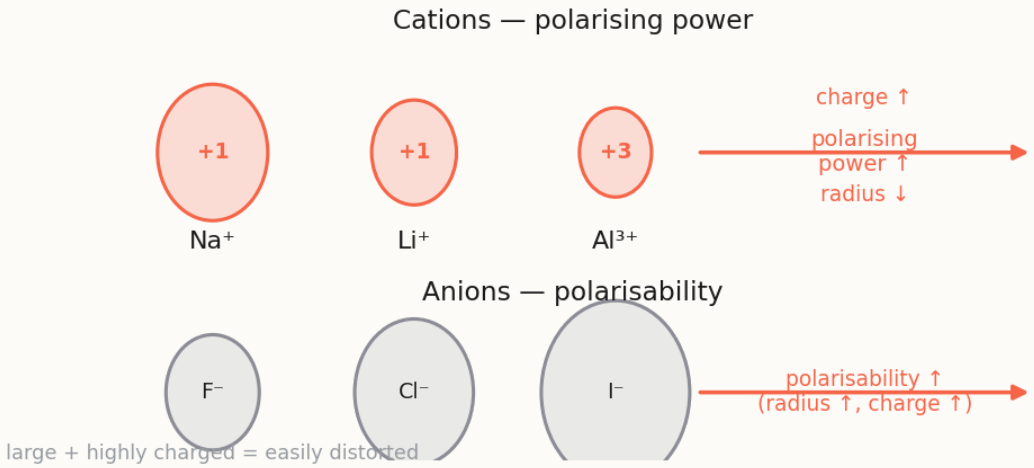


FIG — *Polarising power grows as cations get smaller and more highly charged; polarisability grows as anions get larger and more negatively charged.*

EXAM ALERT

Extension: MgO vs Al_2O_3 melting points

Al_2O_3 (2072 °C) melts **lower** than MgO (2852 °C) despite Al^{3+}'s higher charge — the strong polarisation in Al_2O_3 introduces covalent character, weakening the purely ionic attraction. A classic application of 3.8 and 3.9.

SPEC 3.10 ≈ 25 MIN

The Covalent Bond

THE SPECIFICATION SAYS

understand that covalent bonding is the strong electrostatic attraction between two nuclei and the shared pair of electrons between them, based on the evidence:

Shared pairs, sigma and pi bonds — and the evidence that proves them.

DEFINITION

Covalent bond

The strong electrostatic attraction between the two nuclei of the bonded atoms and the shared pair of electrons held between them.

- When a covalent bond is formed, two atomic orbitals overlap and a **molecular orbital** is formed.
- Like atomic orbitals, a molecular orbital can only contain **two electrons**.
- Overlap of atomic orbitals is thus only possible if **both orbitals contain one electron each** (a normal covalent bond), or if **one orbital is full and the other is empty** (a dative covalent bond).
- Atoms can share more than one pair of electrons and form double or triple covalent bonds:

- **A single bond** results from one shared pair of electrons — e.g. in chlorine, Cl_2 (Cl—Cl).
- **A double bond** results from two shared electron pairs — e.g. in oxygen, O_2 (O=O).
- **A triple bond** results from three shared electron pairs — e.g. in nitrogen, N_2 (N≡N).

Sigma (σ) bonds — the single covalent bond

- For a single covalent bond, an orbital containing one electron overlaps **linearly (head-on)** with an orbital containing one electron of the other atom.
- There can only be **one sigma bond between two atoms**.
- **s–s overlap** — formation of hydrogen, H_2: the two 1s orbitals overlap head-on.
- **s–p overlap** — e.g. in hydrogen bromide, HBr: the hydrogen 1s orbital overlaps with the bromine p orbital containing the single electron.
- **p–p overlap** — formation of fluorine, F_2: the half-filled p orbitals of the two atoms overlap head-on.

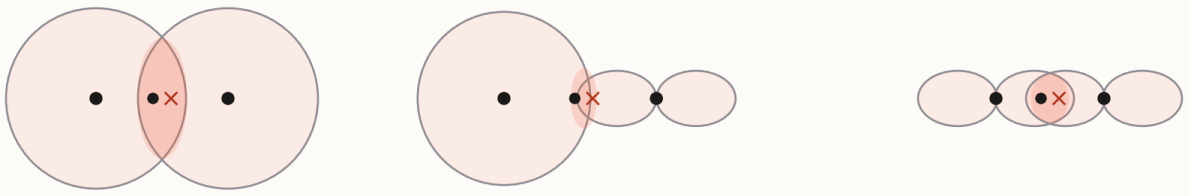


FIG – *A sigma bond forms by linear (head-on) overlap of two half-filled orbitals — s–s (H_2), s–p (HBr) and p–p (F_2).*

Pi (π) bonds — double and triple bonds

- When a double bond is formed between two atoms, one bond is a **sigma bond** and the other is a **pi bond**.
- A pi bond is formed when two p orbitals overlap **laterally (sideways)**.
- Each p orbital has two lobes — one above and one below the plane of the nuclei.
- The bonding region of a pi bond lies **above and below the two nuclei**, not directly between them.
- The sideways overlap of two p orbitals to form a pi bond is **less effective** than the direct, head-on overlap that forms a sigma bond — so a **pi bond is weaker than a sigma bond**.
- **A double bond = one σ + one π** (e.g. ethene, C_2H_4); **a triple bond = one σ + two π** (e.g. ethyne, H—C≡C—H).

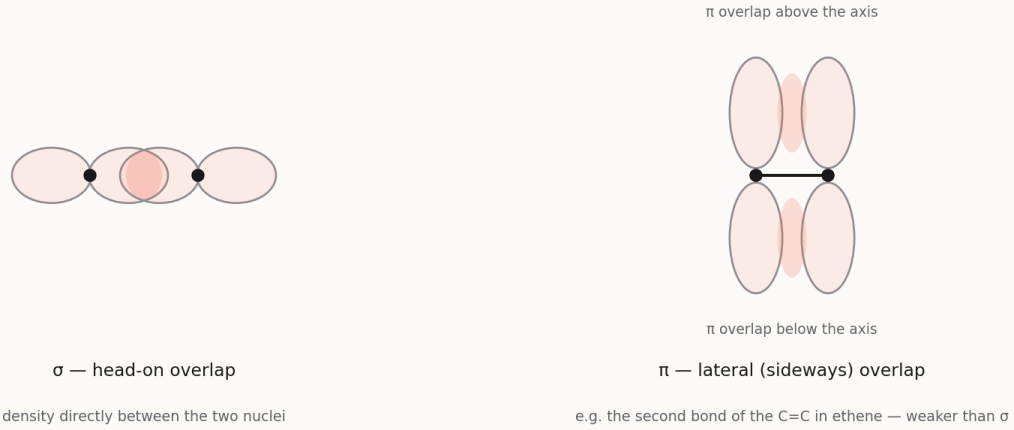


FIG – *A pi bond forms by lateral overlap of p orbitals — the shared density sits above and below the internuclear axis, making it weaker than a sigma bond.*

Bond length and bond strength

DEFINITION

Bond length

The distance between the nuclei of two atoms that are covalently bonded together.

- The strength of a covalent bond is measured by the **amount of energy required to break one mole of the bonds in the gaseous state**.

- **Shorter bonds are stronger bonds** — the closer the nuclei, the greater the electrostatic attraction between the two nuclei and the electrons in the overlapping atomic orbitals.

Evidence 1 — the physical properties of giant atomic structures

- The physical properties of giant atomic structures (also called **giant molecular structures**), such as diamond and graphite, provide evidence for the strong electrostatic attraction in covalent bonding.
- They are **very hard and have high melting temperatures** — the structures are made up of a large number of strong covalent bonds, all of which need to be broken if the atoms are to be separated.
- They **do not conduct electricity** as they have no mobile ions or electrons — but **graphite conducts**: each atom uses only three of its outer-shell electrons for bonding, and the remaining electron can move through the layers, allowing conduction.
- They are **insoluble** in water and other solvents.

Evidence 2 — electron density maps

- Electron density maps show **high electron density between atoms that are covalently bonded** — the shared pair of electrons sits between the two nuclei.

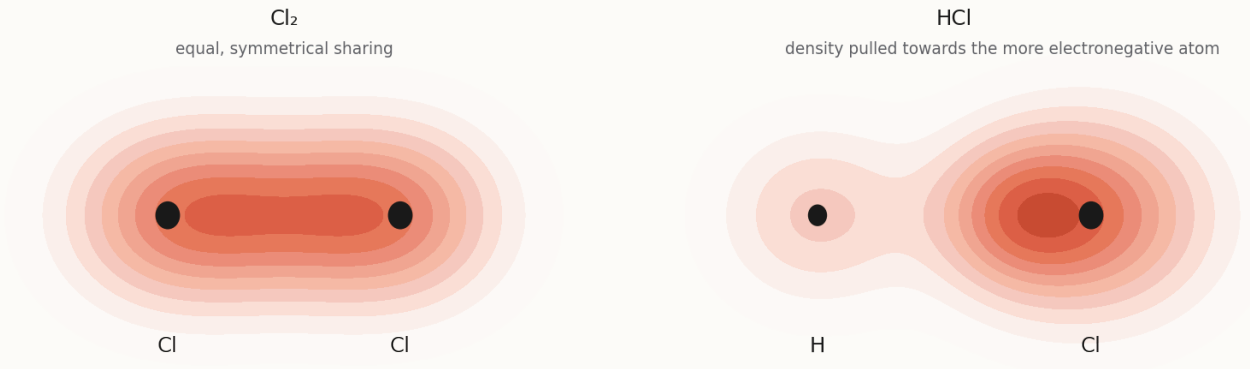


FIG — *Electron density maps for Cl_2 and HCl — the region of high electron density between the two nuclei is the shared pair that constitutes the covalent bond.*

SPEC 3.11 ≈ 25 MIN

Dot-and-Cross Diagrams

THE SPECIFICATION SAYS

be able to draw dot-and-cross diagrams to show electrons in covalent substances, including:

Outer-shell bookkeeping: single, double, triple and dative covalent bonds on paper.

Dot-and-cross diagrams for discrete molecules

DEFINITION

Discrete molecule

An electrically neutral group of two or more atoms held together by covalent bonds.

- Use **dots** for the electrons from one of the atoms and a **cross** for the electrons from the other atom.
- If there are more than two types of atom, use additional symbols such as a small circle or a small triangle.
- Draw the **outer electrons in pairs** to emphasize the number of **bond pairs** and the number of **lone pairs** (non-bonded electrons).

DEFINITION

Lone pair

An electron pair that is part of an atom's valence electron shell but is not involved in bonding.

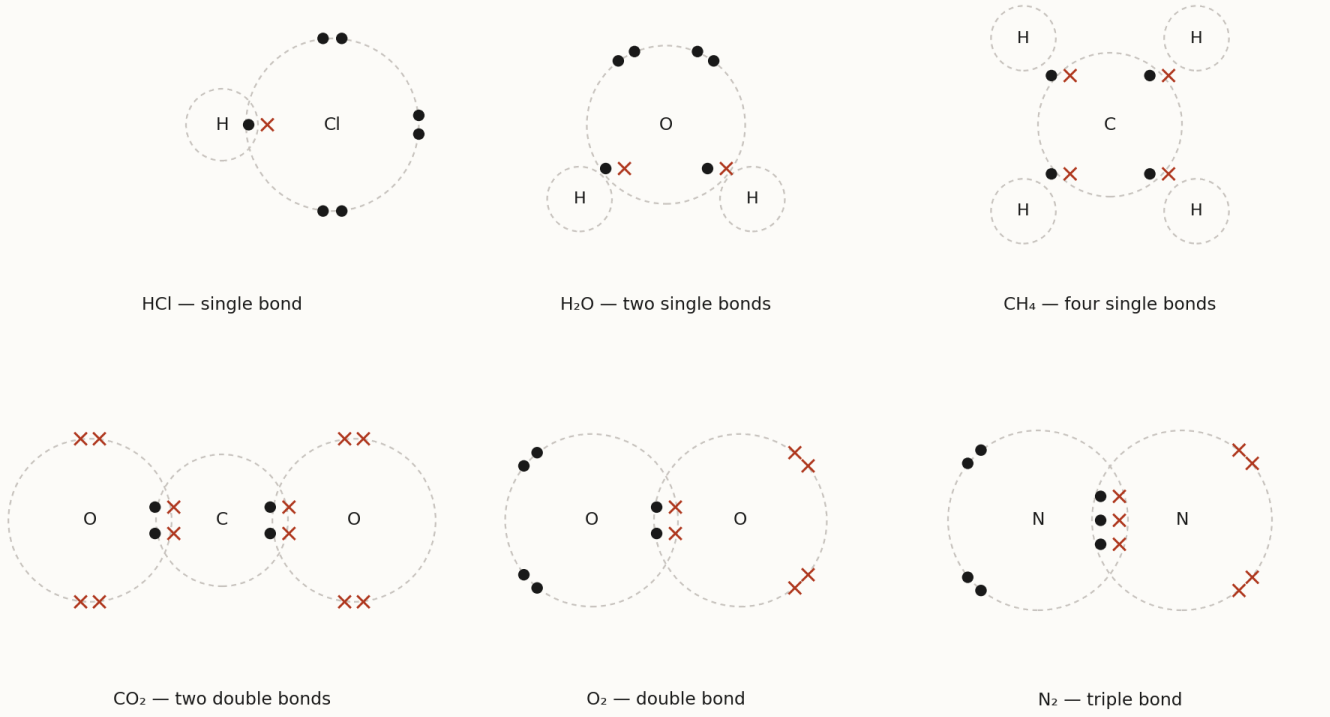


FIG – *Outer-shell dot-and-cross diagrams — single bonds (HCl, H_2O, CH_4), a double bond (CO_2 and O_2, two shared pairs) and a triple bond (N_2, three shared pairs).*

- A **displayed formula** (full structural formula) shows each bonding pair with a line drawn between the two atoms.

The octet rule

- In most cases, covalently bonded molecules have the same number of electrons in the outer shell as a noble gas — mostly **8 electrons**, referred to as the **octet rule**.
- But this is not always true — some species are exceptions, like **$BeCl_2$, BCl_3, PCl_5 and SF_6**.

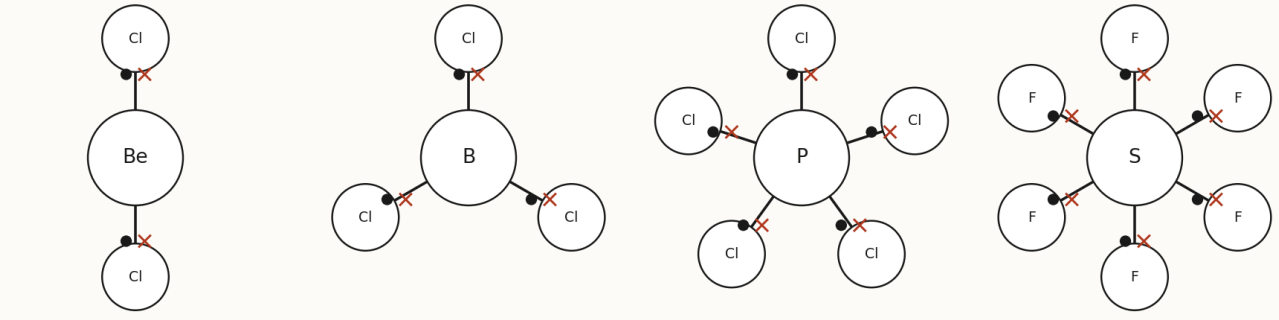


FIG – *The four classic exceptions to the octet rule — beryllium and boron are electron-deficient; phosphorus and sulfur expand their octets.*

COMPOUND	ELECTRON PAIRS AROUND THE CENTRAL ATOM	ELECTRONS AROUND THE CENTRAL ATOM
Beryllium chloride, $BeCl_2$	2 bond pairs, 0 lone pairs	4
Boron trichloride, BCl_3	3 bond pairs, 0 lone pairs	6
Phosphorus(V) chloride, PCl_5	5 bond pairs, 0 lone pairs	10
Sulfur hexafluoride, SF_6	6 bond pairs, 0 lone pairs	12

Dative covalent (coordinate) bonds

DEFINITION

Dative covalent (coordinate) bond

A covalent bond formed when **both of the shared electrons come from just one of the atoms**.

- It is formed when an orbital of one atom containing **two electrons (a lone pair)** overlaps with an **empty orbital** of the other atom.
- The basic requirement: one atom with a **lone pair** of electrons (non-bonded) and the other atom having an orbital with **no electrons**.
- The **ammonium ion, NH_4^+** — the nitrogen of NH_3 donates its lone pair into the empty 1s orbital of an H^+ ion.
- The **hydroxonium ion, H_3O^+** — the oxygen of H_2O donates its lone pair to an H^+ ion.
- The **aluminium chloride dimer, Al_2Cl_6** — two electron-deficient $AlCl_3$ molecules combine: a chlorine lone pair on each $AlCl_3$ donates into the empty orbital of the aluminium of the other molecule, giving two dative bonds.

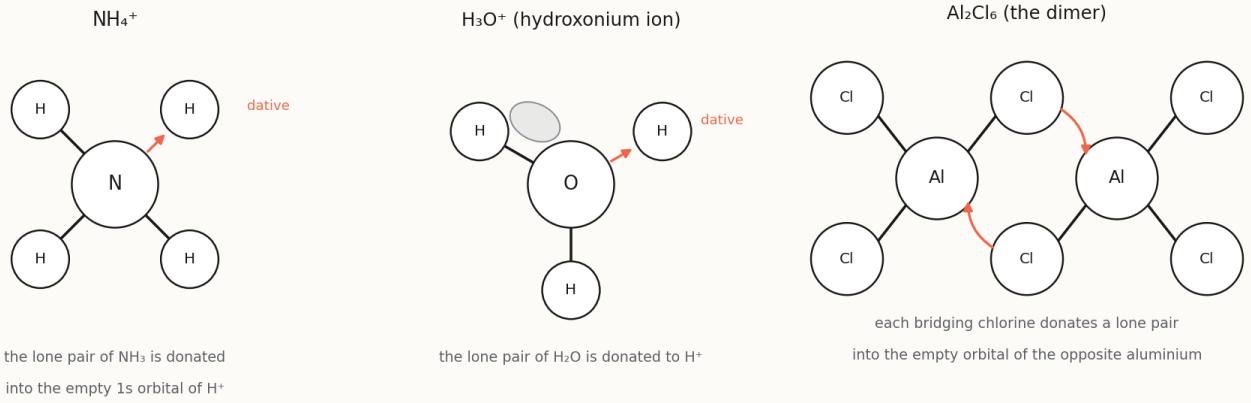


FIG — *Dative covalent bonds — the lone pair of NH_3 completes the shell of H^+ (NH_4^+); the lone pair of H_2O does the same (H_3O^+); in Al_2Cl_6 each bridging chlorine donates a lone pair to the opposite aluminium.*

KEY POINT

Once formed, a dative bond is an ordinary covalent bond

The arrow head only records **where the electrons came from**. A dative covalent bond has the same strength and the same length as any other covalent bond — in NH_4^+ all four N—H bonds are identical.

SPEC 3.12 ≈ 25 MIN

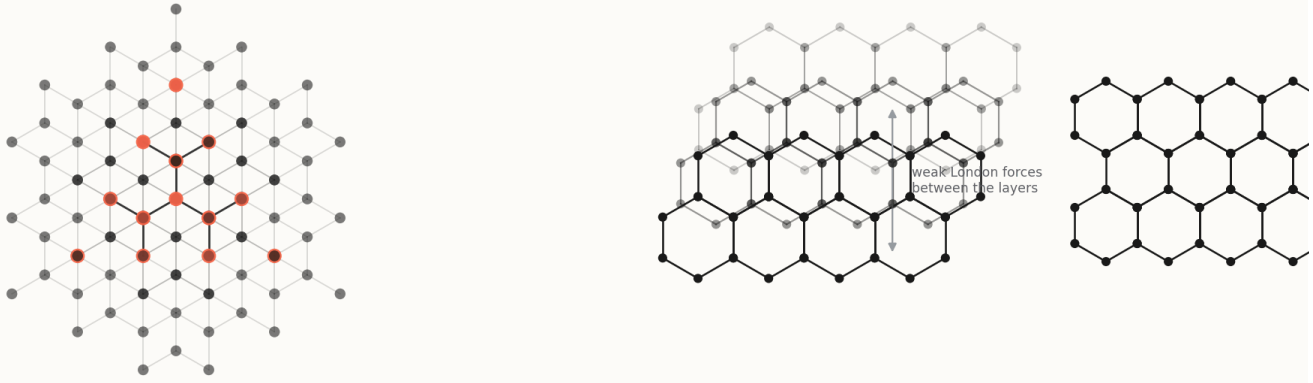
Giant Covalent Lattices

> THE SPECIFICATION SAYS
>
> be able to describe the different structures formed by giant lattices of carbon atoms, including graphite, diamond and graphene, and discuss the applications of each

Diamond, graphite and graphene — and how the four solid structures compare.

Giant covalent lattices

- Giant covalent lattices are sometimes called **network covalent lattices** — they consist of a giant network of atoms linked to each other by covalent bonds.
- The most common giant covalent substances are **diamond**, **graphite** and **graphene**.



Diamond — tetrahedral 3D network

Graphite — layers of hexagons

Graphene — one atom thick

FIG — *The three giant lattices of carbon — diamond: a 3D tetrahedral network; graphite: layers of hexagons held by weak London forces; graphene: a single one-atom-thick honeycomb sheet.*

Diamond

- Composed of **interlocking tetrahedral** carbon atoms.
- Each carbon atom forms **four sigma bonds** to four other carbon atoms in a giant three-dimensional tetrahedral arrangement.
- Diamond is **extremely hard** because of the very strong C—C bonding throughout the structure.
- It also has a **very high melting temperature**, because a great number of strong C—C bonds have to be broken in order to melt it — this requires a large amount of energy.
- Its uses are based on its exceptional hardness: used in **drilling, cutting and grinding**, and as **bearings**.
- It can be made artificially by subjecting graphite to **high temperatures and pressures**.

Graphite

- Composed of **planes of trigonal carbon atoms** joined in a honeycomb pattern.
- Each carbon atom is bonded to **three other carbon atoms at 120°**.

- **Weak London forces (van der Waals forces)** exist between the layers.
- Used as a **lubricant** — it is slippery as the layers slide over each other.
- Used as the **'lead' in pencils**, as an **electrical conductor** and **electrode material** in dry cells, and in **nuclear reactors**.
- It **can conduct electricity** as there are free electrons — one in each atom, since each carbon uses only three of its four outer electrons for bonding.

Graphene

- Graphene is pure carbon in the form of a **very thin sheet, one atom thick**.
- The carbon atoms are arranged in a **honeycomb-like pattern**, bonded in exactly the same way as in graphite — it can therefore be described as a **one-atom-thick layer of graphite**.
- It is **transparent, flexible and highly conductive**.
- A sheet of graphene can be rolled into a ball to produce **fullerene molecules**, and into a cylinder to produce **carbon nanotubes**.
- Graphene is **very strong**, so it can be used to enhance the strength of other materials — research has demonstrated that adding even a trace amount of graphene to plastics, metals or other materials can make these materials much stronger.
- Graphene is **highly conductive**, hence it could be useful in microelectronics (for example, to make **LED lighting** more efficient and longer lasting) and also in larger applications — for example **thermal foils for mobile devices**.

Structure and properties — the four types of solid structure

The four types of solid structures covered in this topic are **giant metallic lattices**, **giant ionic lattices**, **giant covalent lattices** and **discrete (simple) molecular lattices**.

- **Giant metallic lattices** — a regular arrangement of positive metal ions (cations) surrounded by a sea of mobile delocalised electrons. Fairly high to high melting and boiling temperatures; good thermal and electrical conductivity; malleable and ductile.
- **Giant ionic lattices** — a regular arrangement of positive ions (cations) and negative ions (anions). Fairly high melting and boiling temperatures; brittle; poor electrical conductivity when solid but conducts when molten or aqueous; often soluble in water.
- **Giant covalent lattices** — a giant network of atoms linked to each other by strong covalent bonds. High to very high melting and boiling temperatures; extremely hard; non-conductors (graphite is the exception); insoluble.
- **Molecular solids** — collections of molecules held together by **intermolecular forces (IMFs)**. The solid structure is maintained by intermolecular forces rather than by bonds (metallic, ionic or covalent). Generally low melting and boiling temperatures: melting requires only the weak intermolecular forces to be broken, which is less than the energy needed to break covalent bonds.

BONDING	STRUCTURE	EXAMPLES
Metallic	Giant lattice	Mg, Al, Cu, Zn, Na

BONDING	STRUCTURE	EXAMPLES
Ionic	Giant lattice	NaCl, MgO, CsF
Covalent (including polar covalent)	Giant lattice	C (diamond), C (graphite), Si, SiO_2 (sand), BN
Covalent (including polar covalent)	Molecular	H_2O, I_2, P_4, S_8, C_{60}, $C_{12}H_{22}O_{11}$ (sucrose)
Covalent (including polar covalent)	Macromolecular	Polymers e.g. polythene, proteins, DNA

	GIANT METALLIC	GIANT IONIC	GIANT COVALENT	SIMPLE MOLECULAR
Particles present	Positive ions and delocalised electrons	Positive and negative ions	Atoms	Molecules
Type of bonding	Metallic	Ionic	Covalent	Covalent (within the molecules)
Presence of intermolecular forces	No	No	No	Yes
Melting and boiling temperatures	Fairly high to high	Fairly high to high	High to very high	Generally low
Electrical conductivity	Good when solid and molten	Non-conductor when solid; conducts when molten and aqueous	Non-conductor (graphite conducts)	Non-conductor
Solubility in water	Insoluble unless the metal reacts with water (e.g. Na, Li)	Generally soluble — exceptions are AgCl, AgBr, AgI and $BaSO_4$	Insoluble	Generally insoluble, but may dissolve if hydrogen bonding is possible (e.g. sucrose) or if the substance reacts with water (e.g. Cl_2)

SPEC 3.13 ≈ 15 MIN

Electronegativity

THE SPECIFICATION SAYS

understand the meaning of the term ‘electronegativity’ as applied to atoms in a covalent bond

The tug-of-war for the bonding pair, measured on the Pauling scale.

DEFINITION

Electronegativity

The ability of an atom to attract the bonding pair of electrons in a covalent bond to itself.

- Electronegativity is usually measured using the **Pauling scale**.
- **Fluorine** is the most electronegative element according to this scale — it is given a value of **4.0**.
- **Caesium** has the lowest value, **0.7**.

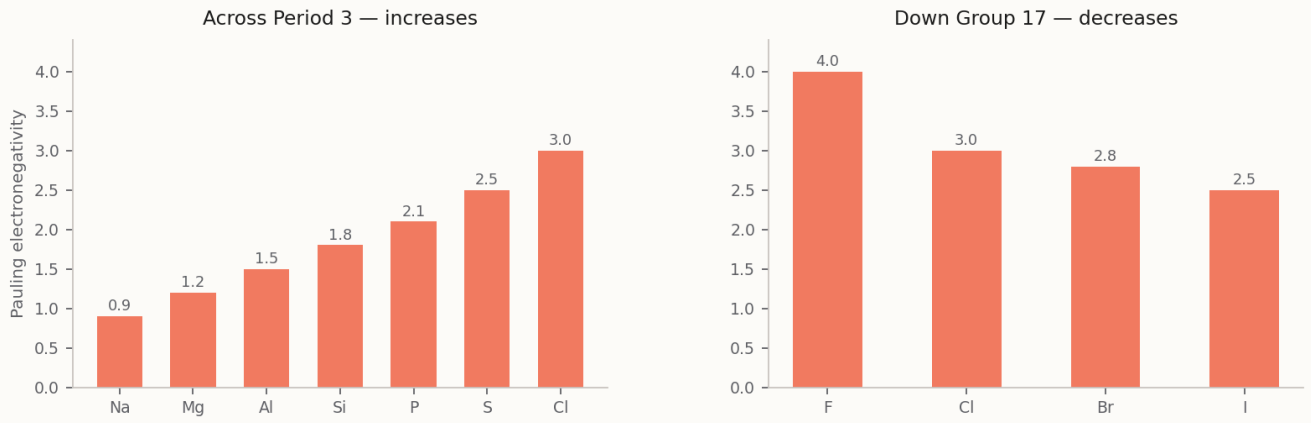


FIG — *Pauling electronegativity values — rising across Period 3 (Na 0.9 → Cl 3.0) as nuclear charge increases, and falling down Group 17 (F 4.0 → I 2.5) as shielding increases.*

Trends in electronegativity

- **Electronegativity increases across a period**, because:
- Nuclear charge increases across the period.
- The outer electrons are in the same shell, so there is no extra shielding.
- Therefore, there is more attraction between the nucleus and the shared pair of electrons.

- **Electronegativity decreases down a group**, because:
- The outer electrons are further away from the nucleus.
- There is increased shielding.
- Therefore, there is less attraction between the nucleus and the shared pair of electrons.

SPEC 3.14 ≈ 20 MIN

The Bonding Continuum

> THE SPECIFICATION SAYS
>
> **know that ionic and covalent bonding are the extremes of a continuum of bonding type and be able to explain this in terms of electronegativity differences, leading to bond polarity in bonds and molecules, and to ionic bonding if the electronegativity is large enough**

From pure covalent to ionic — the electronegativity difference decides where a bond sits.

Most bonds are not purely ionic or purely covalent but somewhere in between — **intermediate bonding**. Electronegativity is the main factor which leads to this intermediate bonding.

Distribution of electron density in a covalent bond

- **Non-polar bond** — similar atoms have the same electronegativity, so they both pull on the electrons to the same extent: the electrons are **equally shared**, the electron density is symmetrical, and the bond is a **non-polar (pure) covalent bond**.
- **Polar covalent bond** — when the bonded atoms differ in electronegativity they share the electron pair **unequally**, giving a partial negative charge (**δ^-**) on the more electronegative atom and a partial positive charge (**δ^+**) on the less electronegative atom.
- The atom with the higher electronegativity pulls the electron pair closer to its end, so the electron density is unsymmetrical.
- A **dipole** is induced and the bond is said to be **polar** — the greater the electronegativity difference, the greater the bond polarity.
- A polar covalent bond can be represented using an **arrow** to show the direction of electron drift (pointing towards the δ^- end).

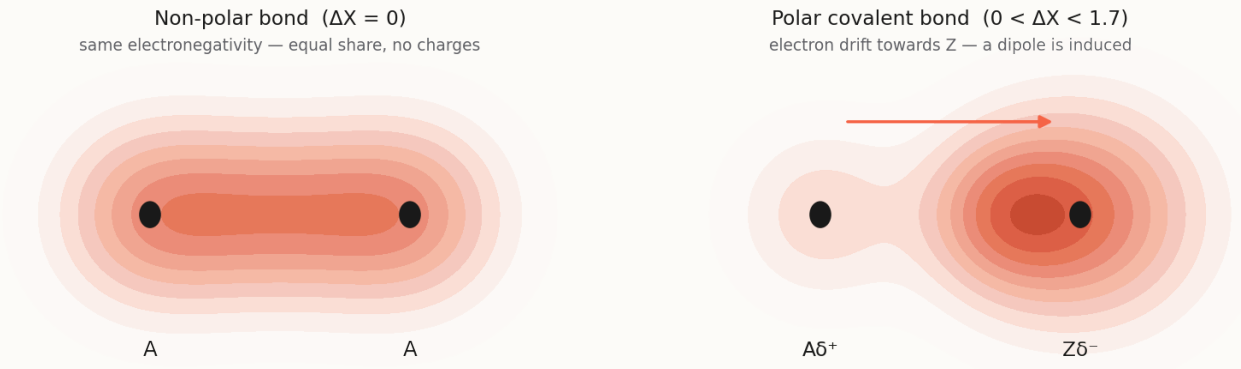


FIG — *Equal sharing gives a non-polar bond; unequal sharing gives δ^+/δ^- ends and a dipole — the arrow points in the direction of electron drift.*

Using electronegativity to classify bonds

- The ionic and covalent bondings are the **two extremes of bonding** — in practice, all degrees of intermediate bonding occur, having both ionic and covalent characteristics.
- The degree of ionic and covalent character depends on the **electronegativity difference (ΔX)** between the two atoms.
- It is somewhat artificial to classify bonds purely on ΔX, but these ranges are useful:

ELECTRONEGATIVITY DIFFERENCE ΔX	BOND TYPE	CHARGES
$\Delta X \geq 1.7$	Ionic — predominantly ionic with some covalent character	Full charges: A^+ and Z^-
$0 < \Delta X < 1.7$	Polar covalent — predominantly covalent with some ionic character	Partial charges: $A\delta^+$ and $Z\delta^-$
$\Delta X = 0$	Pure covalent — the atoms share the electrons equally	No charges

WORKED EXAMPLE

Classifying bonds from electronegativity differences

PROBLEM

Classify the bonding in NaCl, LiF, $SrBr_2$, H_2O, H_2 and Cl_2. (Pauling values: F 4.0, Cl 3.0, Br 2.8, O 3.0, Na 0.9, Li 1.0, Sr 1.0, H 2.1)

1. **NaCl:** $\Delta X = 3.0 - 0.9 = 2.1 \rightarrow \Delta X \geq 1.7$, so the bond is **ionic**.
2. **LiF:** $\Delta X = 4.0 - 1.0 = 3.0 \rightarrow$ **ionic**.
3. **$SrBr_2$:** $\Delta X = 2.8 - 1.0 = 1.8 \rightarrow$ **ionic**.
4. **H_2O:** $\Delta X = 3.0 - 2.1 = 0.9 \rightarrow 0 < \Delta X < 1.7$, so the bonds are **polar covalent**.
5. **H_2 and Cl_2:** $\Delta X = 0$ (identical atoms), so the bonds are **pure covalent**.

ANSWER **NaCl, LiF and $SrBr_2$ are ionic (ΔX = 2.1, 3.0, 1.8); H_2O is polar covalent (ΔX = 0.9); H_2 and Cl_2 are pure covalent (ΔX = 0).**

KEY POINT

Water is neither purely ionic nor purely covalent

Water does not contain positive and negative ions, and its electrons are not shared equally. Water is best described as a **polar compound**: one end (or pole) of the molecule carries a partial positive charge (δ^+) and the other end a partial negative charge (δ^-).

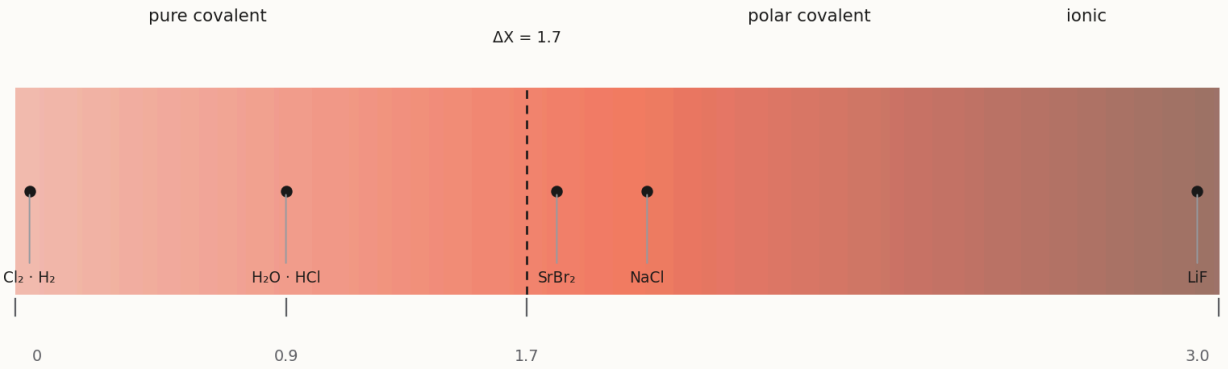


FIG – *The bonding continuum — ionic and covalent are the extremes; the electronegativity difference decides where a real bond sits between them.*

SPEC 3.15 ≈ 20 MIN

Polar Bonds and Polar Molecules

> THE SPECIFICATION SAYS
>
> be able to distinguish between polar bonds and polar molecules and predict whether or not a given molecule is likely to be polar

Polar bonds don't guarantee a polar molecule — symmetry decides.

- **Polarity is a vector quantity** — it has both magnitude and direction.
- Some molecules that contain polar bonds are polar — the molecule has a **permanent dipole**.
- Some molecules may **not** be polar even if they contain polar bonds.
- Molecules will be polar if they have a **net dipole moment** — it is a bit like balanced forces:
- In a **non-polar molecule**, the dipoles of the bonds within the molecule **cancel each other**.
- In a **polar molecule**, the dipoles of the bonds within the molecule **do not cancel each other**.
- In **symmetrical molecules**, though the bonds are polar, the dipoles from each of the bonds cancel out, resulting in no overall polarity — these molecules are **non-polar**.
- Therefore **linear, trigonal planar, tetrahedral, trigonal bipyramidal and octahedral** molecules are non-polar (provided that similar atoms are bonded to the central atom).

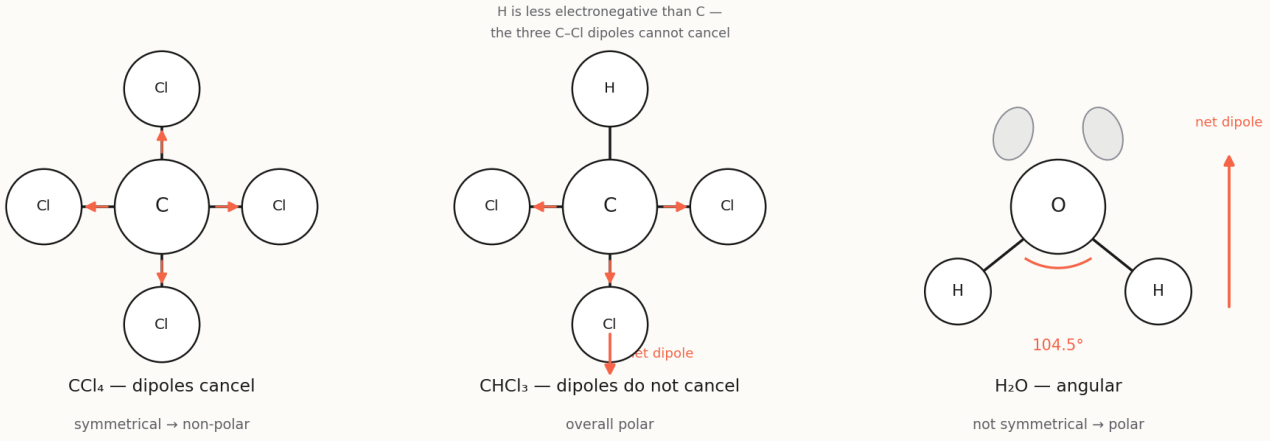


FIG – *CCl_4: four identical bond dipoles cancel — non-polar. $CHCl_3$ and H_2O: the dipoles cannot cancel, so the molecules are polar.*

Polar bonds vs polar molecules

- In a simple molecule like **HCl**, the H—Cl bond is polar because the electronegativity difference is large enough — and since HCl contains only that one bond, the HCl molecule is also described as polar.
- In **tetrachloromethane, CCl_4**, each bond is polar — however, the molecule as a whole is **not polar**. In other words, it does not have an end (or a side) which is slightly negative and one which is slightly positive: there is no overall separation of charge from top to bottom, or from left to right.
- In **trichloromethane, $CHCl_3$**, the hydrogen at the top of the molecule is less electronegative than carbon and so is slightly positive — the molecule now has a slightly positive 'top' and a slightly negative 'bottom', and so is overall a **polar molecule**.

- The electrons in the trichloromethane molecule are attracted towards one end of the molecule — this distortion of the electron cloud means that the centre of positive charge and the centre of negative charge **do not occur in the same place** in the molecule.

- **Angular and pyramidal molecules are polar.**
- **Example — water, H_2O:** oxygen is more electronegative than hydrogen, so it has a greater pull on the shared electrons; the oxygen is slightly negative (δ^-) and the hydrogens are slightly positive (δ^+). Because the molecule is not symmetrical, it has an overall dipole moment and is therefore a **polar molecule**.

WATCH OUT

Exception — nitrogen trifluoride, NF_3

NF_3 has the same geometry as NH_3, but the molecule has **very little polarity** (it is often treated as non-polar) because the lone pair cancels out the polarity of the N—F bonds.

Experiment — determining the polarity of molecules

- Place a liquid in a burette and allow a narrow stream (jet) to run out.
- Place a **charged rod** next to the flow of the liquid.
- **Polar molecules** are attracted and deflected towards the charged rod; a jet of non-polar molecules continues straight down.
- Determining the effect of an electrostatic force on jets of **water, ethanol and cyclohexane** therefore decides whether the molecules are polar or non-polar.

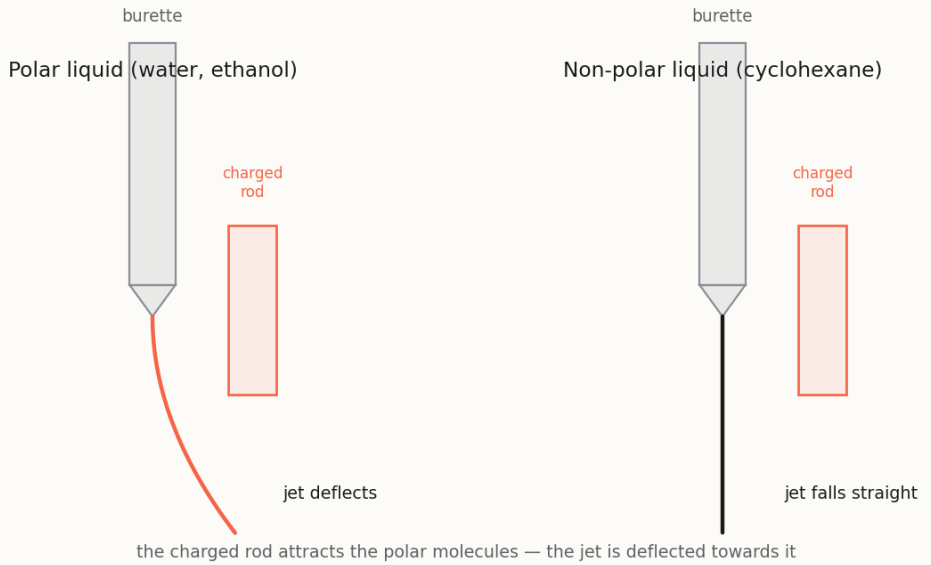


FIG — *The electrostatic force on liquid jets — polar liquids (water, ethanol) are deflected towards a charged rod; non-polar liquids (cyclohexane) are not.*

SPEC 3.16 ≈ 15 MIN

Electron-Pair Repulsion Theory

THE SPECIFICATION SAYS

understand the principles of the electron-pair repulsion theory, used to interpret and predict the shapes of simple molecules and ions

Electron pairs repel — count them and the shape follows.

DEFINITION

Valence Shell Electron Pair Repulsion (VSEPR) theory

The theory used to predict the shapes of simple molecules and ions by considering the repulsions between the pairs of electrons — bond pairs and lone pairs — in the valence shell of the central atom.

- The shape of a molecule or ion is largely determined by the **total number of electron pairs in the valence shell of the central atom**.
- The electron pairs arrange themselves **as far apart as possible** to minimize the repulsion (maximum separation).
- The order of repulsive strength: **lone pair–lone pair > lone pair–bond pair > bond pair–bond pair**.

KEY POINT

The guiding principle

The shape adopted is the one which keeps repulsive forces to a **minimum**.

- To determine a shape, count up the number of **covalent bond pairs** and **lone pairs** around the central atom, and work out the shape which keeps the bonds as far apart as possible.

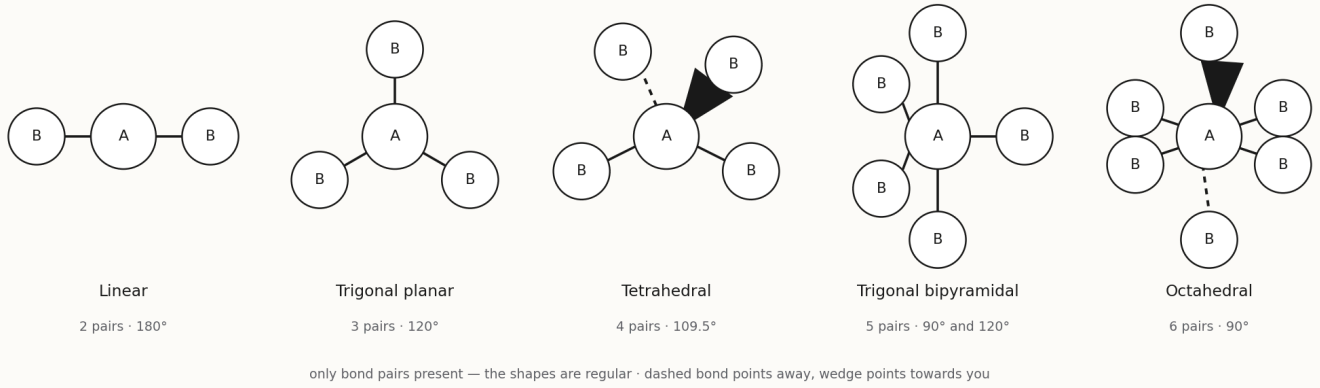


FIG — *The five regular arrangements, with only bond pairs present — 2 pairs: linear 180°; 3: trigonal planar 120°; 4: tetrahedral 109.5°; 5: trigonal bipyramidal 90° and 120°; 6: octahedral 90°.*

SPEC 3.17 ≈ 10 MIN

Bond Length and Bond Angle

THE SPECIFICATION SAYS

understand the terms ‘bond length’ and ‘bond angle’

The two measurements that describe every bond and every angle.

DEFINITION

Bond length

The average distance between the centres of the nuclei of two bonded atoms in a molecule.

DEFINITION

Bond angle

In a molecule made up of three or more atoms, the average angle between the two covalent bonds meeting at the central atom (the angle between the bonded orbitals), written θ.

- A bond angle is only defined where two bonds meet at the same atom — e.g. the H—O—H angle in water is 104.5° and the H—C—H angle in methane is 109.5°.
- Bond length decides how close the two nuclei sit; bond angle decides the shape adopted by three or more atoms.

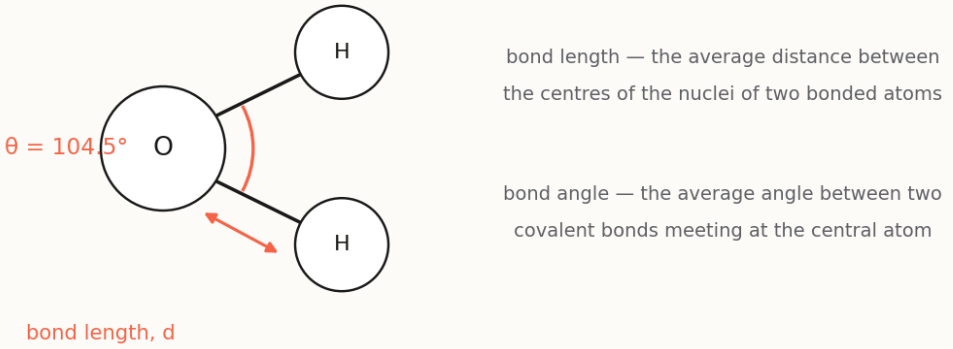


FIG — *The two measurements on one molecule — the bond length d (nucleus to nucleus) and the bond angle θ (between two bonds meeting at oxygen).*

SPEC 3.18 ≈ 25 MIN

Shapes and Bond Angles

> THE SPECIFICATION SAYS
>
> know and be able to explain the shapes of, and bond angles in, $BeCl_2$, BCl_3, CH_4, NH_3, NH_4^+, H_2O, CO_2, gaseous PCl_5, SF_6 and C_2H_4

The ten species you must know — shapes, angles and the reasons behind them.

Species without lone pairs — the regular shapes

- Only bond-pair repulsions occur, so the basic shapes are **regular** — the shape of the molecule depends on the number of bond pairs around the central atom.

SPECIES	BOND PAIRS	LONE PAIRS	SHAPE	BOND ANGLE
$BeCl_2$	2	0	Linear	180°
BCl_3	3	0	Trigonal planar	120°
CH_4	4	0	Tetrahedral	109.5°
NH_4^+	4	0	Tetrahedral	109.5°
PCl_5 (gaseous)	5	0	Trigonal bipyramidal	90° and 120°
SF_6	6	0	Octahedral	90°
CO_2	2 double bonds (act like single bonds)	0	Linear	180°
C_2H_4	3 electron groups around each C (one is the C=C double bond)	0	Trigonal planar around each carbon — the whole molecule is planar	≈120°
NH_3	3	1	Trigonal pyramidal	107°
H_2O	2	2	Angular (bent)	104.5°

Species with lone pairs

- Lone pairs of electrons have **greater repulsive power** than bond pairs, so their presence affects the bond angles — they push the bond pairs away.
- The resulting configuration is based on the **total number of electron pairs**, but the actual shape **does not include the lone pairs** — they are 'invisible' when naming the shape.
- A water molecule is **angular** despite having 4 electron pairs around the oxygen — two of the pairs are lone pairs.
- Lone pairs distort the shape of the molecule and reduce the bond angle: **each lone pair decreases the bond angle by about 2.5°**.

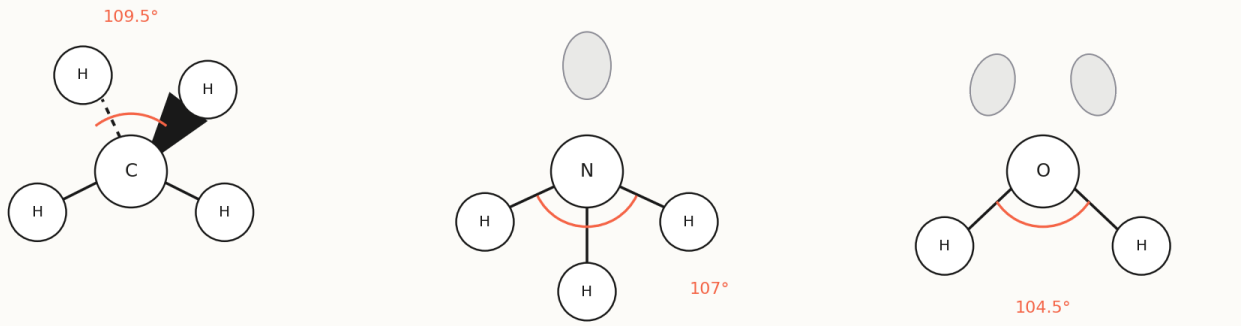


FIG – *The lone-pair compression — four electron pairs give a tetrahedron; each lone pair squeezes the bond angle down by about 2.5°: 109.5° → 107° → 104.5°.*

WORKED EXAMPLE **Ammonia, NH_3**

PROBLEM

Determine the shape of, and the bond angle in, an ammonia molecule.

1. Nitrogen has **3 bond pairs and 1 lone pair** (total = 4 electron pairs).
2. The shape is based on a **tetrahedron**.
3. As the lone pair–bond pair repulsions are greater than the bond pair–bond pair repulsions, the H—N—H bond angle is reduced from 109.5°.

ANSWER **The shape is trigonal pyramidal, with H—N—H = 107°.**

WORKED EXAMPLE **Water, H_2O**

PROBLEM

Determine the shape of, and the bond angle in, a water molecule.

1. Oxygen has **2 bond pairs and 2 lone pairs** (total = 4 electron pairs).
2. The shape is based on a **tetrahedron**.
3. The extra lone pair–lone pair repulsion pushes the H—O—H bond angle down further than in ammonia.

ANSWER **The shape is angular (bent), with H—O—H = 104.5°.**

Species with double bonds

- For repulsive purposes, **double bonds act like single bonds** — each double bond is one region of electron density.

WORKED EXAMPLE **Carbon dioxide, CO_2**

PROBLEM

Determine the shape of, and the bond angle in, a carbon dioxide molecule.

1. Carbon has **2 double bond pairs and no lone pairs**.
2. Since double bonds act like single bonds, the shape is based on two regions of electron density repelling each other — they move as far apart as possible.

ANSWER **The shape is linear, with O—C—O = 180°.**

WORKED EXAMPLE **Sulfur dioxide, SO_2**

PROBLEM

Determine the shape of, and the bond angle in, a sulfur dioxide molecule.

1. Sulfur has **2 double bond pairs and 1 lone pair**.
2. Since the double bonds act like single bonds, the shape is based on two bond pairs and one lone pair repelling each other — a trigonal planar arrangement with one corner occupied by the lone pair.

ANSWER **The shape is angular (bent), with O—S—O = 117.5°.**

SPEC 3.19 ≈ 20 MIN

Predicting Shapes by VSEPR

THE SPECIFICATION SAYS

be able to apply the electron-pair repulsion theory to predict the shapes of, and bond angles in, molecules and ions analogous to those in 3.18

Same method, new species — ions, analogues and organic molecules.

Shapes can be worked out for **ions** and for species analogous to those in 3.18 by the same method. It allows you to predict the shape — but, in some cases, not the true nature of the bonding.

Shapes of simple ions

- Write the **outer electron configuration of the central atom**, adjust for the charge on the ion (a negative charge adds the extra electron to the central atom; a positive charge removes one), then count bond pairs and lone pairs exactly as before.
- **AlH_4^-** — aluminium has 3 outer electrons and the negative charge supplies one more: **4 bond pairs, 0 lone pairs** → tetrahedral, H—Al—H = 109.5°.

- **ICl_2^+** — iodine supplies 2 bond pairs and keeps **2 lone pairs** → angular, based on a tetrahedron, with the angle compressed below 109.5° by the two lone pairs.
- **PF_6^-** — 6 bond pairs, 0 lone pairs → octahedral, F—P—F = 90°.
- **H_3O^+** — the oxygen of water donates its lone pair to H^+: **3 bond pairs, 1 lone pair** → trigonal pyramidal, like NH_3.

WORKED EXAMPLE **The hydroxonium ion, H_3O^+**

PROBLEM

Predict the shape of, and the bond angle in, the hydroxonium ion H_3O^+.

1. Oxygen (Group 16) has 6 outer electrons. Three are used to bond three hydrogen atoms, and the dative bond supplies nothing new to count — the ion has **3 bond pairs and 1 lone pair** (total = 4 electron pairs).
2. The arrangement is based on a **tetrahedron**, with one corner occupied by the lone pair.
3. The lone pair–bond pair repulsion compresses the angle below 109.5° — the same effect as in NH_3.

ANSWER **Trigonal pyramidal, with H—O—H ≈ 107°.**

WORKED EXAMPLE **The tetrahydroaluminate ion, AlH_4^-**

PROBLEM

Predict the shape of, and the bond angle in, the AlH_4^- ion.

1. Aluminium (Group 13) has 3 outer electrons; the negative charge supplies one more — all four form **4 bond pairs** with hydrogen, with **no lone pairs**.
2. Four bond pairs repel equally and arrange themselves in a tetrahedron.

ANSWER **Tetrahedral, with H—Al—H = 109.5°.**

Shapes of organic molecules

- Organic molecules are not given a particular shape name — however, the **3D spatial arrangement** (and the bond angles at each atom) is expected.
- Every **singly-bonded carbon with four electron groups** is tetrahedral, with bond angles of about **109.5°** — e.g. the H—C—H angle in ethane, methanol or chloromethane.
- Every **carbon joined by a double bond (C=C)** is trigonal planar, with bond angles of about **120°** — e.g. the H—C—H angle in ethene, and the C—C=C angle in propene.
- An **—O—H oxygen** carries two bond pairs and two lone pairs, so the C—O—H angle is about **104.5°**, like water.

TIP

Count electron groups, not bonds

For VSEPR a **double bond counts as one group** of electron density. That is why the carbon of C_2H_4 — three groups: two C—H bonds plus one C=C — is trigonal planar (≈120°), while the carbon of CH_4 — four groups — is tetrahedral (109.5°).

SPEC 3.20 ≈ 10 MIN

Metallic Structure

THE SPECIFICATION SAYS

understand that metals consist of giant lattices of metal ions in a sea of delocalised electrons

Cations in fixed layers, wrapped in a sea of delocalised electrons.

DEFINITION

Metallic structure

A giant three-dimensional lattice of positive metal ions arranged in regular layers, surrounded by a sea of mobile, delocalised outer-shell electrons.

- Each metal atom **donates its outer-shell electrons** into a shared 'sea' — sodium donates one per atom, magnesium two, aluminium three.
- Having lost its outer electrons, each atom becomes a **positive ion (cation)**, fixed in position within the regular layered lattice.
- The donated electrons are **delocalised** — they belong to the whole lattice, not to any particular ion — and are free to move throughout the structure.
- The lattice is held together by the attraction between the fixed cations and this mobile sea of electrons — that attraction is **metallic bonding** (3.21).

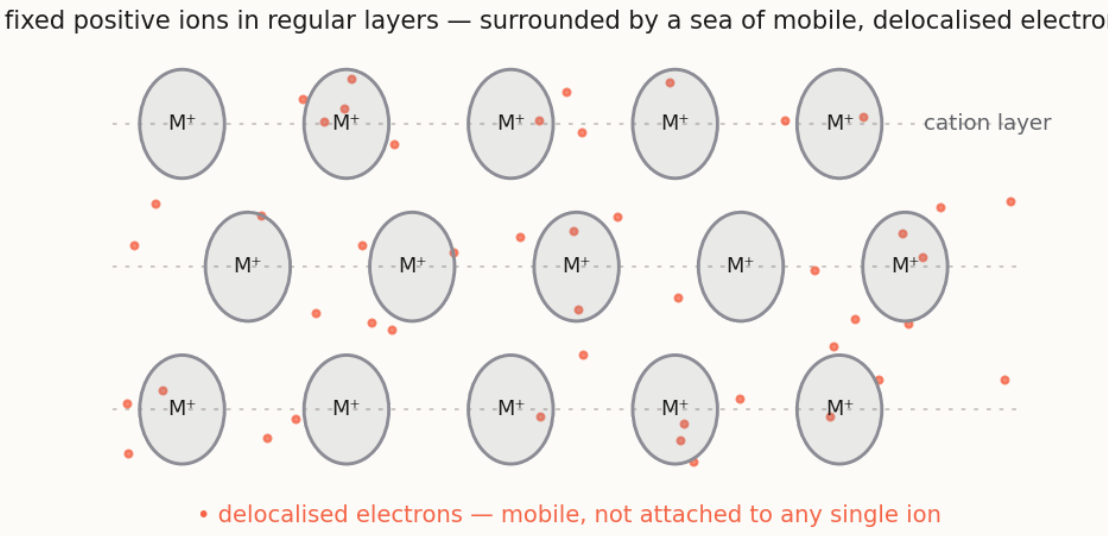


FIG – *The metallic model — fixed cations in regular layers, wrapped in a sea of delocalised electrons that moves freely through the whole lattice.*

KEY POINT

'Delocalised' is the key word

The outer electrons are no longer attached to any single atom. This mobility is the source of every property explained in 3.22 — conductivity in particular.

SPEC 3.21 ≈ 10 MIN

Metallic Bonding

THE SPECIFICATION SAYS

know that metallic bonding is the strong electrostatic attraction between metal ions and the delocalised electrons

The strong electrostatic attraction between metal ions and delocalised electrons.

DEFINITION

Metallic bonding

The strong electrostatic attraction between the positive metal ions (cations) and the sea of delocalised electrons.

- Every cation is attracted to **the whole shared sea of electrons**, and the attraction acts **in all directions** — there are no individual 'bonds' between pairs of atoms.
- The strength of the attraction rises with the **charge of the cations** and falls with their **radius** — the same charge-density idea as ionic bonding (3.6).

- It also rises with the **number of delocalised electrons donated per atom** — aluminium (three electrons per atom) has stronger metallic bonding than sodium (one).
- This is why sodium melts at just 98 °C, while magnesium (650 °C) and aluminium (660 °C) need far more energy.

TIP

Spot the difference from ionic bonding

Ionic: attraction between **oppositely charged ions**. Metallic: attraction between **cations and shared delocalised electrons**. The electron sea is not made of anions — the electrons have no fixed position at all.

SPEC 3.22 ≈ 15 MIN

Properties of Metals

THE SPECIFICATION SAYS

be able to use the models in 3.20 and 3.21 to interpret simple properties of metals, including electrical conductivity and high melting temperature

Conductivity, melting temperature and malleability — all explained by the model.

The value of the metallic model is how neatly it explains the everyday behaviour of metals — the specification asks for electrical conductivity and high melting temperature in particular.

High melting and boiling temperatures

- The strong electrostatic attraction between the cations and the delocalised electrons requires **large amounts of thermal energy** to overcome.
- The melting temperature **increases with cation charge** — $Al^{3+} > Mg^{2+} > Na^{+}$ — because more highly charged ions attract the sea more strongly.
- It also increases as the ionic radius falls and as **more delocalised electrons are donated per ion**, both of which strengthen the bonding.

METAL	ION	ELECTRONS DONATED PER ATOM	MELTING POINT
Sodium	Na^{+}	1	98 °C
Magnesium	Mg^{2+}	2	650 °C
Aluminium	Al^{3+}	3	660 °C

Electrical conductivity

- Metals are excellent conductors **in both the solid and the liquid state** — the delocalised electrons are mobile whatever the cations are doing.
- When a potential difference is applied, the electron sea **flows through the structure**, carrying charge from one end to the other.

- Contrast the ionic case (3.1): ionic solids cannot conduct because their ions are fixed — metals have **mobile electrons even in the solid**.

Malleability and ductility

- The layers of cations can **slide over one another** when a force is applied — the shape changes instead of shattering.
- The metallic bond survives the slide: the **non-directional sea of delocalised electrons adjusts** to the new arrangement and keeps holding the layers together.
- This is why metals can be **hammered into sheets (malleable)** and **drawn into wires (ductile)**.

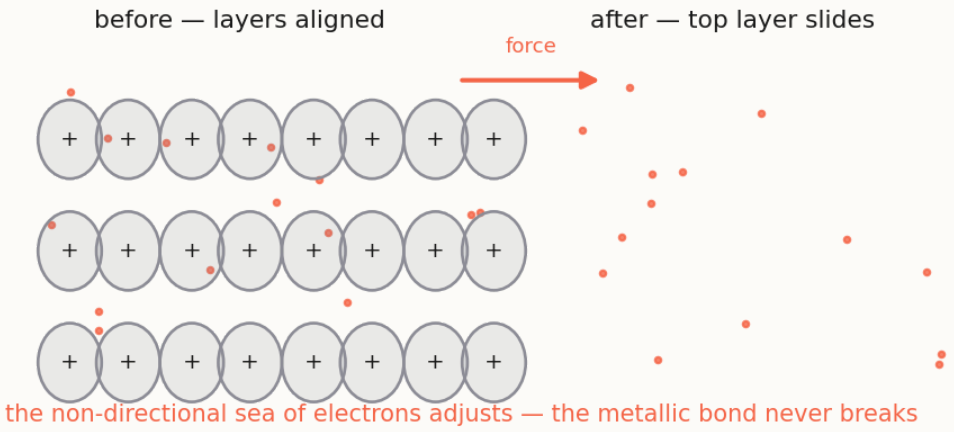


FIG – *Why metals bend rather than break — the cation layers slide, and the non-directional electron sea simply adjusts to the new shape.*

WORKED EXAMPLE Soft sodium, harder magnesium

PROBLEM

Use the metallic bonding model to explain why sodium is soft with a melting point of 98 °C while magnesium is harder and melts at 650 °C.

1. Sodium: each atom donates **one electron**, and the Na^+ ions carry a **single positive charge**.
2. Magnesium: each atom donates **two electrons**, and the Mg^{2+} ions are **doubly charged** and slightly smaller.
3. The attraction between cations and the electron sea is therefore much stronger in magnesium.

ANSWER **Stronger metallic bonding in magnesium means more energy is needed to separate the ions — hence its higher melting point and greater hardness.**

EXAM ALERT

Malleable metal vs brittle ionic crystal

When a force makes layers slide in a **metal**, the electron sea adjusts and the structure stays intact. In an **ionic solid**, sliding brings like-charged ions face to face — the repulsion shatters the crystal. Same force, opposite outcome.

END OF TOPIC 03

01 · Formulae, Equations and Amount of Substance

02 · Atomic Structure and the Periodic Table

04 · Introductory Organic Chemistry and Alkanes

05 · Alkenes

06 · Energetics

07 · Intermolecular Forces

08 · Redox Chemistry and Groups 1, 2 and 7

09 · Introduction to Kinetics and Equilibria

10 · Organic Chemistry: Halogenoalkanes, Alcohols and Spectra

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